DIABETES PREDICTION USING LOGISTIC REGRESSION

Machine Learning Project Summary

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Domain	AI-ML	**Dataset**	Pima Indians Diabetes

1. Objective

To build a Logistic Regression classification model that predicts whether a person is likely to have diabetes based on selected medical measurements.

2. Dataset and preprocessing

The Pima Indians Diabetes dataset contains medical measurements such as pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function and age. The target variable is **Outcome** (0 = non-diabetic; 1 = diabetic). Zero values in Glucose, BloodPressure, SkinThickness, Insulin and BMI were treated as missing values. Missing values were imputed using the median, and features were standardized. The data was divided into training and testing sets using an 80:20 stratified split (random_state = 42).

3. Exploratory data analysis

Exploratory analysis was performed using outcome counts, feature distributions, correlation analysis and boxplots. These visualizations helped examine class distribution, feature patterns and possible outliers.

4. Model and evaluation

A Logistic Regression model was trained on the processed training data. At a probability threshold of 0.40, the test-set confusion matrix was:

	Predicted 0	**Predicted 1**
Actual 0	77	23
Actual 1	20	34

Metric	**Result**
Accuracy	72.08%
Precision (class 1)	59.65%
Recall (class 1)	62.96%
F1-score (class 1)	61.26%

5. Conclusion

The Logistic Regression model was able to identify diabetic and non-diabetic cases from the given medical features, achieving an accuracy of about 72.08% on the test set. Using a 0.40 threshold improved the detection of diabetic cases compared with the 0.50 threshold, although some non-diabetic cases were classified as diabetic. This shows that the model can provide a useful baseline prediction, but it is not perfect and should not be used as a substitute for medical diagnosis. Its performance could be improved by testing other algorithms and validating the model on additional data.

Dataset source: Kaggle – Pima Indians Diabetes Dataset